

0 1 . 1

In the following passage, each numbered space can be filled with a biological term.

Each type of cell has specific molecules on its _____ (1) _____ that identify it. An antibody is a _____ (2) _____ with two _____ (3) _____ binding sites. Antibodies are made and released by _____ (4) _____ cells.

Vaccines can provide protection for individuals and populations against disease. The response to a vaccine is an example of _____ (5) _____ immunity.

_____ (6) _____ immunity occurs when a large proportion of a population becomes immune to a disease.

Write the correct biological term beside each number below that matches the space in the passage.

[3 marks]

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

0 1 . 2

There is currently no effective vaccine available for HIV.

Suggest **one** reason why.

[1 mark]

0 1 . 3

Table 1 shows the action of two antibiotics.

Table 1

Antibiotic	Action of antibiotic
Ciprofloxacin	Causes double-stranded DNA to break apart
Penicillin	Prevents formation of crosslinks between murein chains

Using the information in **Table 1**, explain why each antibiotic is **not** effective against HIV.

[2 marks]

Ciprofloxacin _____

Penicillin _____

Turn over for the next question

Turn over ►

